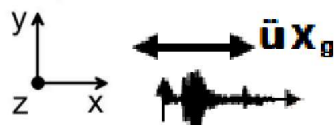
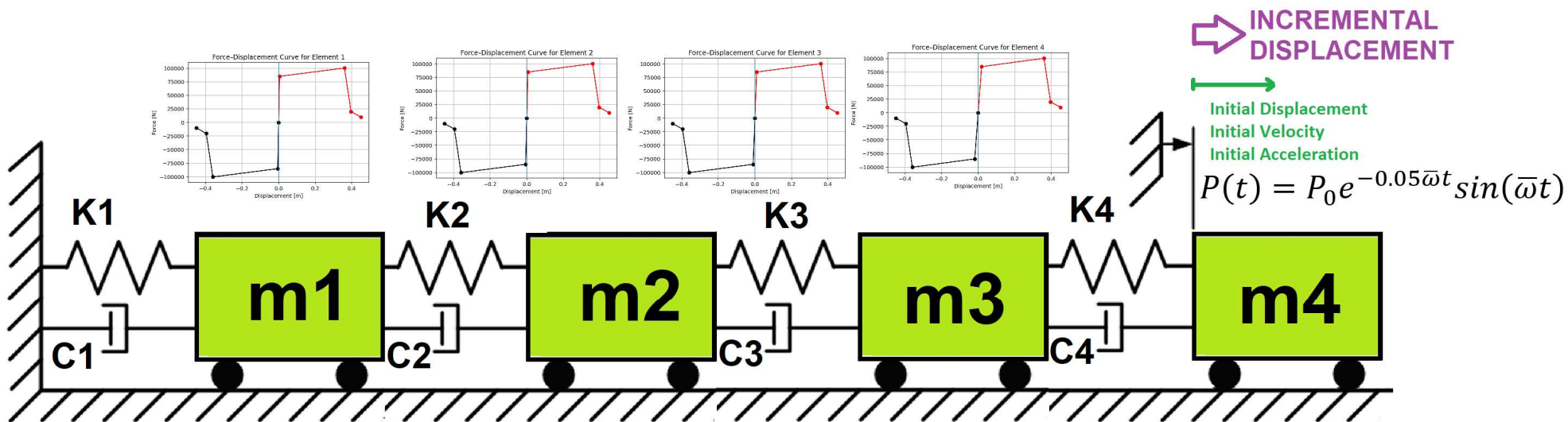


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

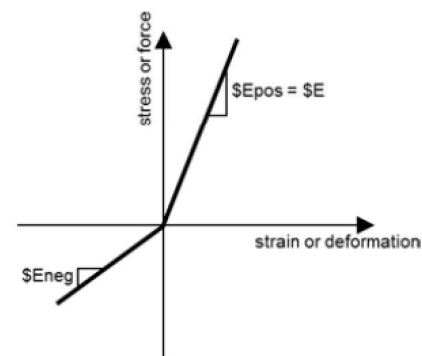
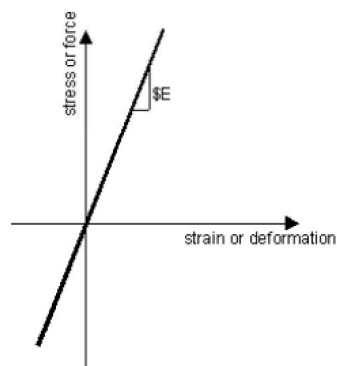
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

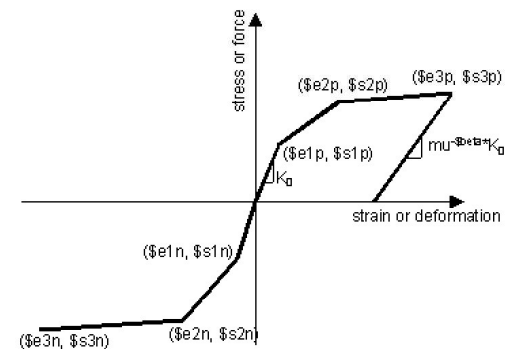
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

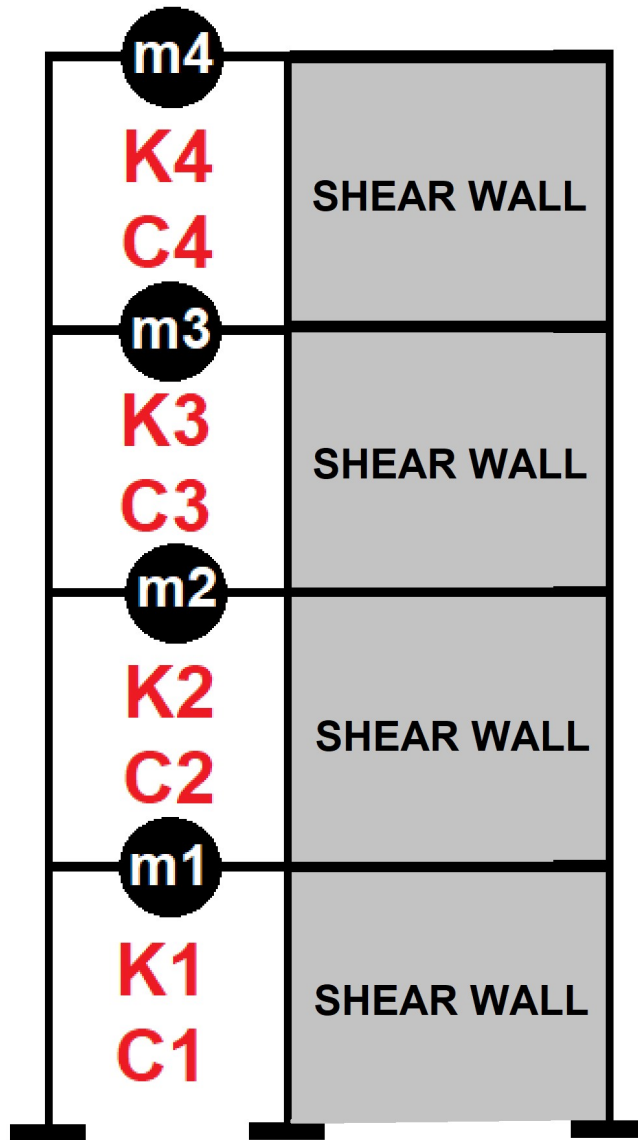
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



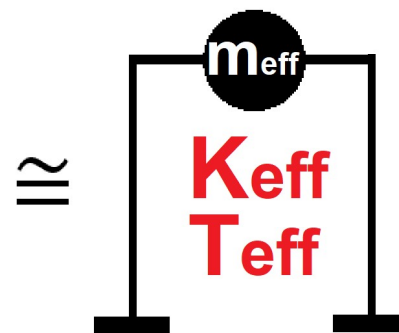
ELASTIC MATERIAL



INELASTIC MATERIAL

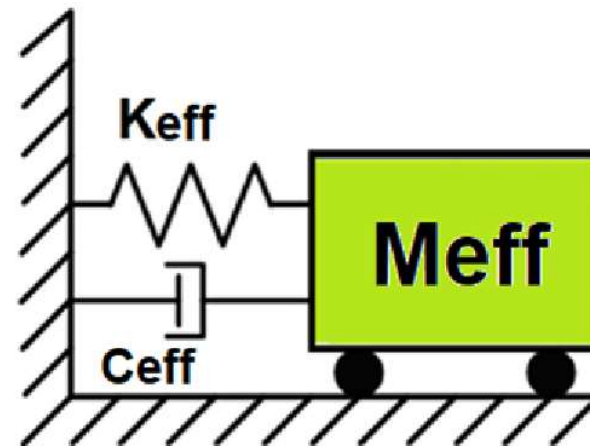


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment

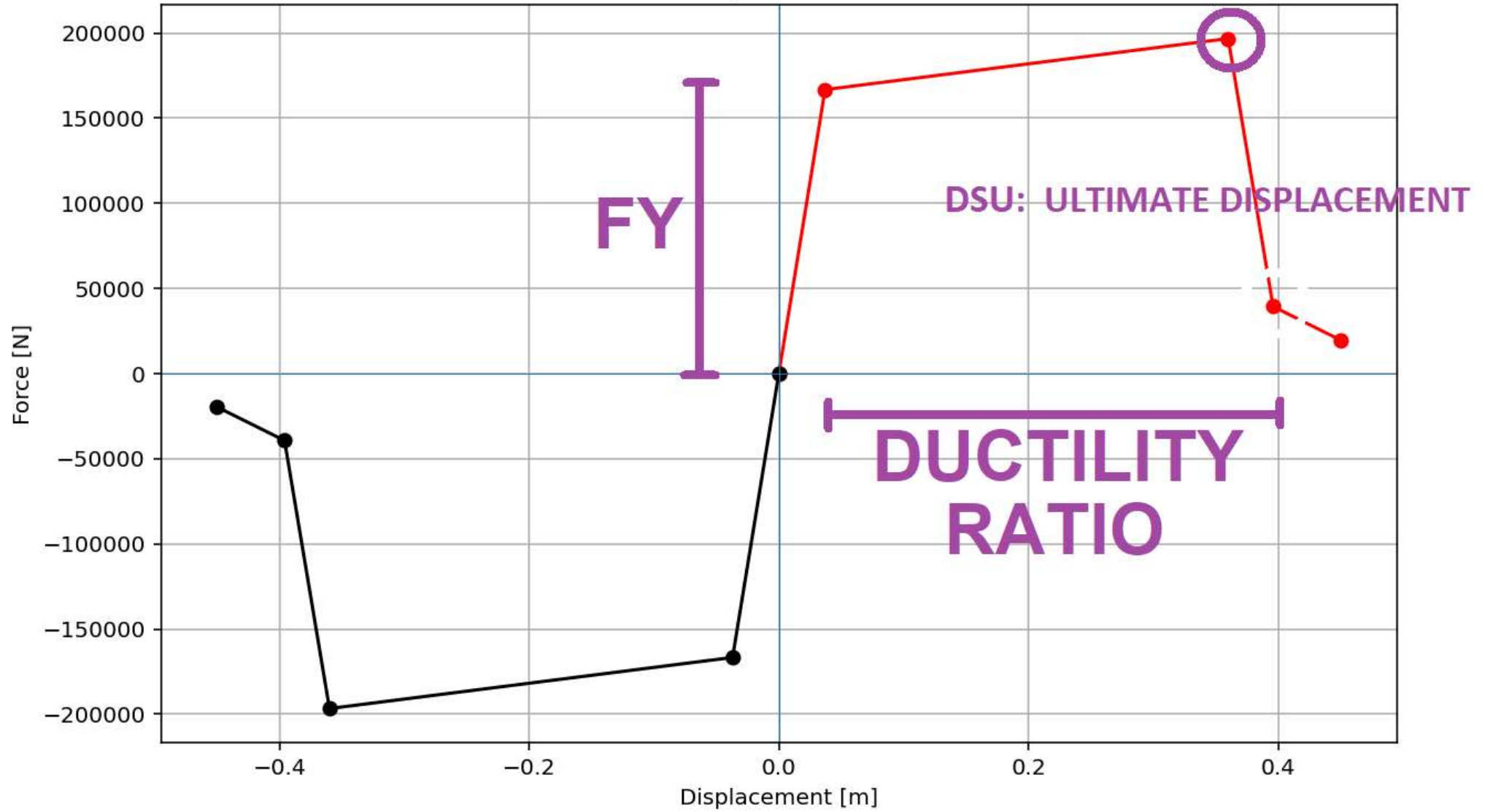


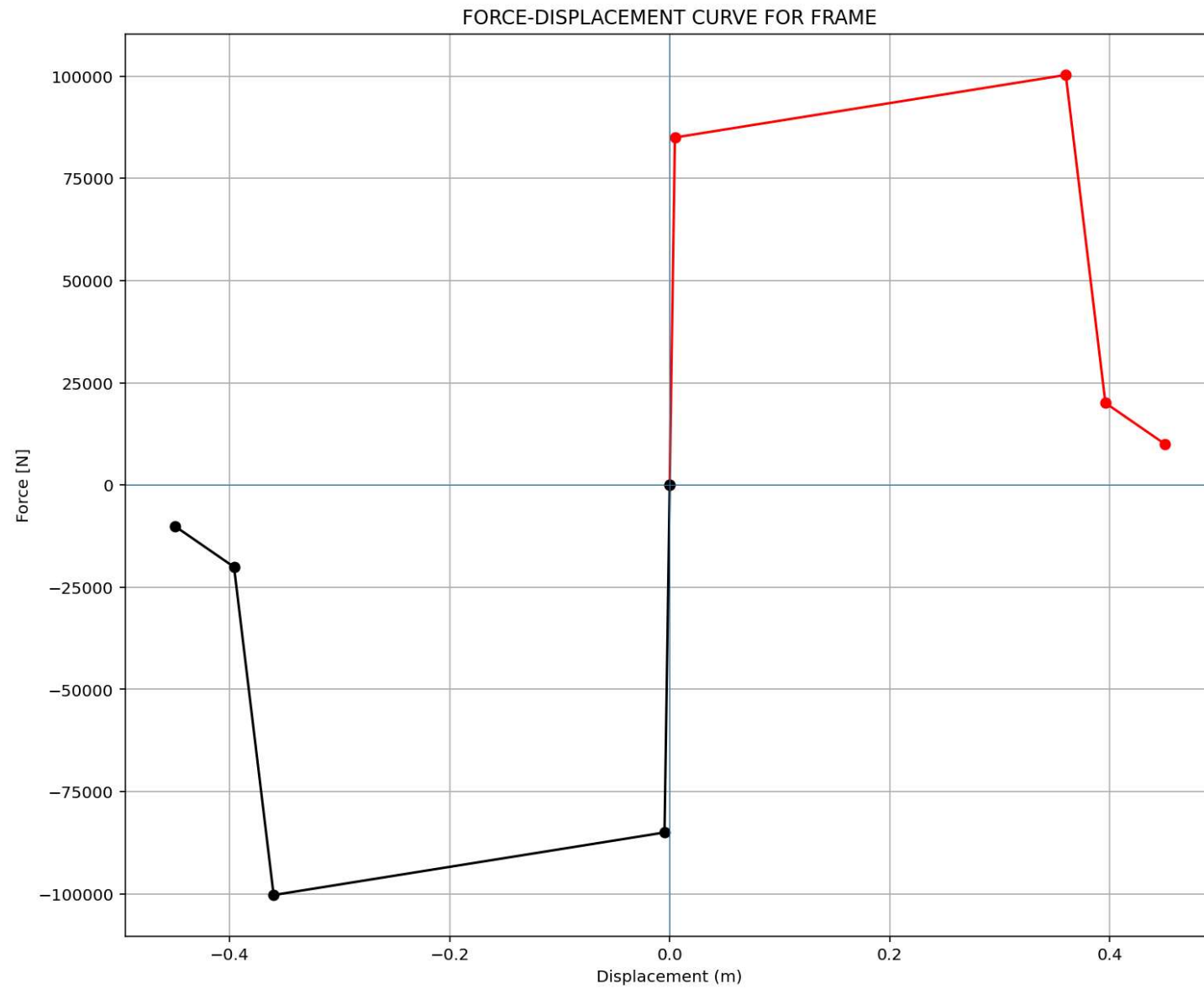
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

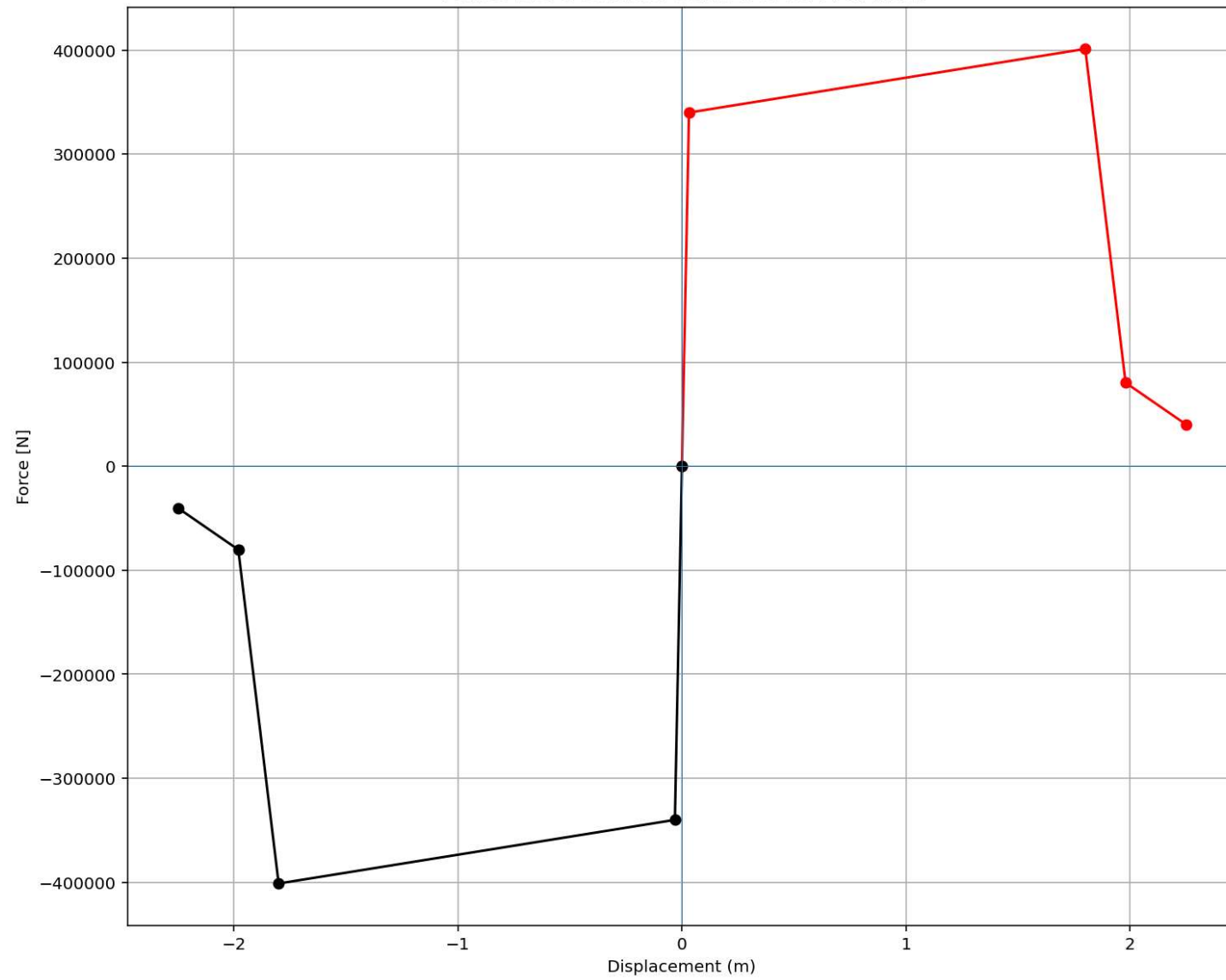


Force-Displacement Curve





FORCE-DISPLACEMENT CURVE FOR SHEAR-WALL



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

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P(t)_MDOF_TWO_DUCT...TION_DECI_8_ANA.py X DISSIPATED_ENERGY_WITH_PLOT_FUN.py X

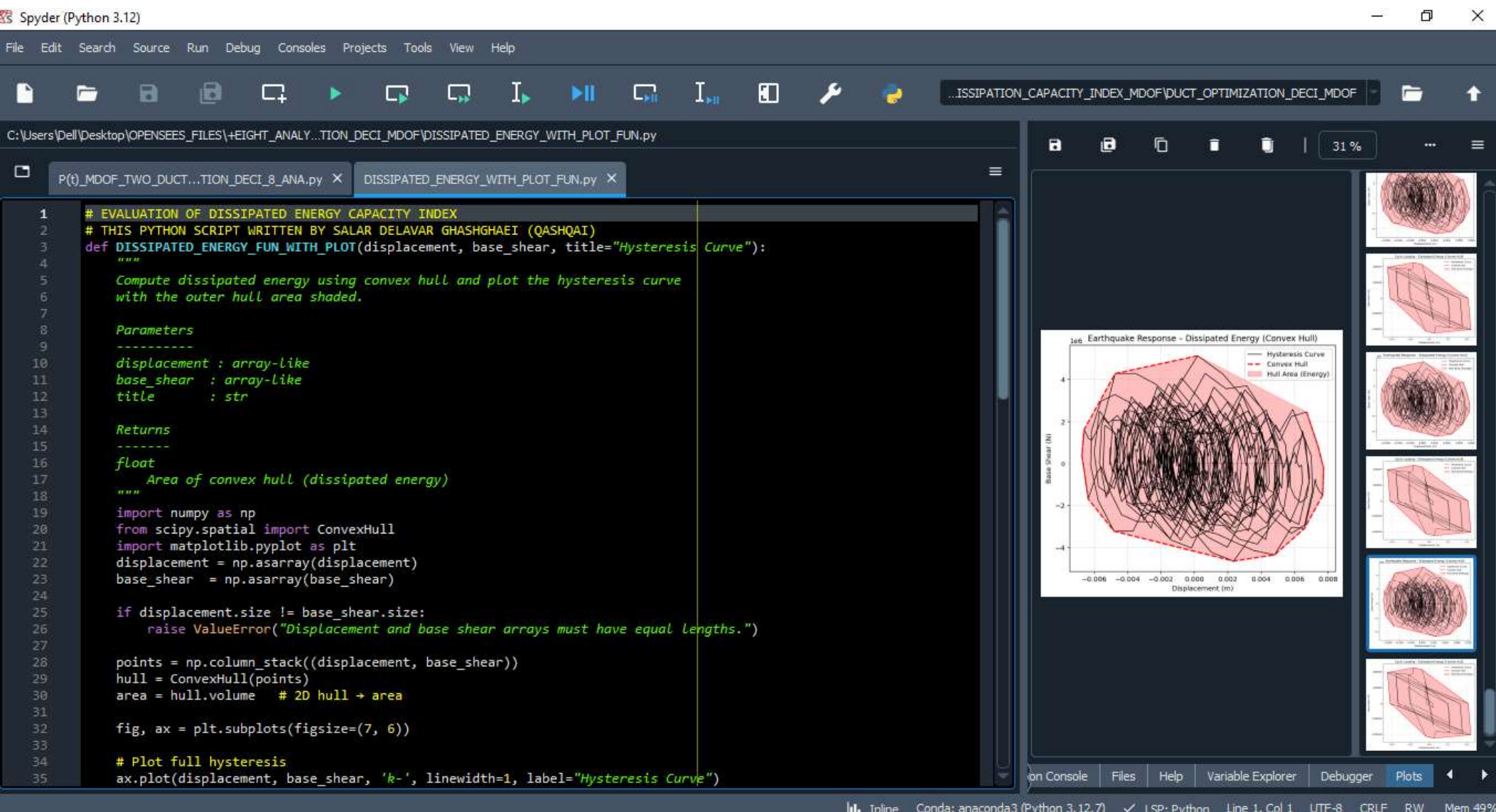
```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL << #
3 # DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY INDEX USING #
4 # FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES #
5 #-----#
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM STRUCTURE : AN OPENSEES FRAMEWORK #
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS #
8 #-----#
9 # EACH STORY HAS FRAME AND SHEAR-WALL (OR BRACE) INTERACTION #
10 #-----#
11 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALYSIS#
12 #-----#
13 # P(t) = P0 sin(wt) #
14 # P(t) = P0 exp(-0.05wt) sin(wt) #
15 #-----#
16 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION #
17 # OF ENERGY DISSIPATION CAPACITY INDEX #
18 #-----#
19 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI) #
20 # EMAIL: salar.d.ghashghaei@gmail.com #
21 #####
22 """
23 Nonlinear Seismic Performance Assessment of a MDOF:
24 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Loading
25
26 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
27 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
28 (elastic-perfectly plastic or hysteretic steel with strain hardening)
29 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
30 and large displacements-essential for collapse assessment.
31
32 Eight analysis protocols are implemented:
33 (1) [PERIOD] : Structural Period
34 (2) [STATIC] : Gravity load analysis establishing dead load state
```

31 %

Earthquake Response - Dissipated Energy (Convex Hull)

on Console Files Help Variable Explorer Debugger Plots

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 167, Col 22 UTF-8 CRLF RW Mem 49%



OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...DOF\P(t)_MDOF_TWO_DUCT_OPTIMIZATION_DECI_8_ANA.py

P(t)_MDOF_TWO_DUCT...TION_DECI_8_ANA.py

```
996 TOTAL MASS = 5000.0 # [kg] Total Mass of Structure
997 %%%-----
998 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
999 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1000
1001 # -----
1002 # FIND THE OPTIMUM VALUE (NEWTON-RAPHSON SOLVER FOR OPTIMAL DUCTILITY RATIO)
1003 # -----
1004 import time as TI
1005 import numpy as np
1006
1007 X = 4.0 # Initial Guess for X -> Ductility Ratio of Structure
1008 ESP = 1e-3 # Finite difference derivative Convergence Tolerance
1009 TOLERANCE = 1e-6 # Convergence Tolerance
1010 RESIDUAL = 100 # Convergence Residual
1011 IT = 0 # Initial Iteration
1012 ITMAX = 100000 # Max. Iteration
1013 DEMAND = 20.0 # [%] Target Value -> ENERGY DISSIPATION CAPACITY INDEX
1014
1015 # Analysis Durations:
1016 starttime = TI.process_time()
1017
1018 while (RESIDUAL > TOLERANCE):
1019     # X -----
1020     ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1021     DATA = MDOF_TWO(X, MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1022
1023     (reaction_CP, disp_CP, DI_CP, _, _,
1024      PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
1025
1026     ANAL_TYPE = 'SEISMIC'
1027     DATA = MDOF_TWO(X, MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1028     (time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI, _, _
```

Console 1/A

```
+-----+
| lambda | omega | period | frequency |
+-----+
| 1.463e+04 | 120.9375 | 0.0520 | 19.2478 |
| 7.177e+04 | 267.9069 | 0.0235 | 42.6387 |
+-----+
```

Time: 20.0000, Displacement: 0.0001 mm, Reaction: -136.94 N

SEISMIC ANALYSIS DONE.

Dissipated Energy from Earthquake= 107144.43 N·m
Dissipated Energy from Cyclic Displacement= 535745.79 N·m

DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]

ZONE 2: MINOR DAMAGE

Fmax: -0.0008826609401708652
DF: -0.8827349793687489
DX: -5.5438256264300047e-08
IT: 11 - RESIDUAL: 5.5438256264300047e-08 X: 23.402375631762727

Optimum X (DUCTILITY RATIO): 23.4024
Iteration Counts: 11
Convergence Residual: 5.5438256264e-08

Total time (s): 84.1406

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 203, Col 13 UTF-8 CRLF RW Mem 48%

Dissipated Energy from Earthquake= 732813.284 N·m
Dissipated Energy from Cyclic Displacement= 3664139.377 N·m

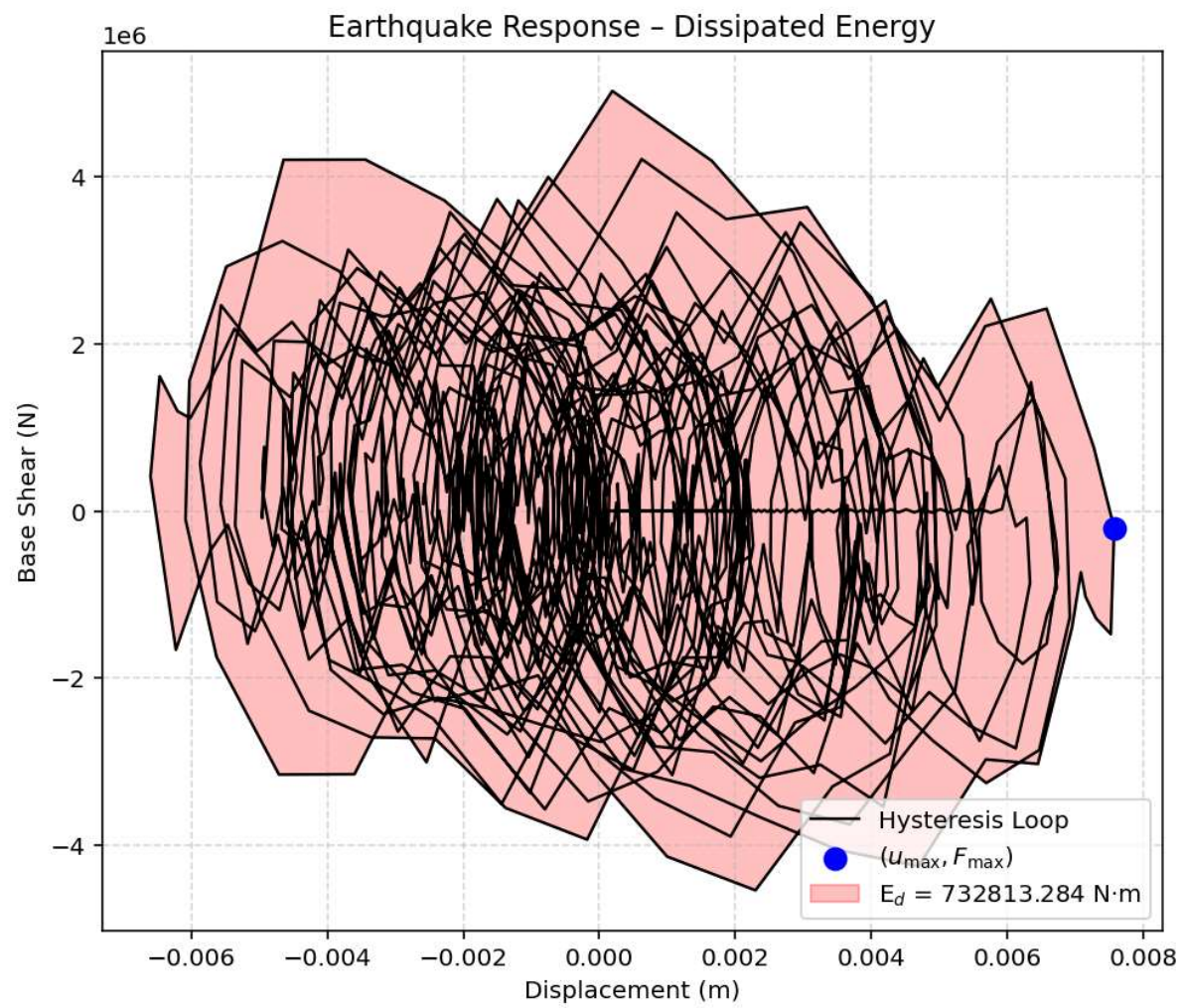
DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]

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Fmax: -0.0008826609401708652
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Optimum X (DUCTILITY RATIO): 23.4024
Iteration Counts: 11
Convergence Residual: 5.5438256264e-08

Total time (s): 84.1406



Cyclic Loading - Dissipated Energy

